SAMBP Technical Report TR-39/2026

Islanding Detection and Protection Mode Switching
for IBR Microgrids

PhD Thesis Supporting Report | 2026

Abstract

This report develops an islanding detection and protection mode switching framework for an IBR microgrid with very low equivalent inertia ($H_{\mathrm{eq}} = 0.5\,\mathrm{s}$). Passive detection via ROCOF ($\geq 1\,\mathrm{Hz/s}$) detects islanding for all power mismatches $|\Delta P| \geq 0.02\,\mathrm{pu}$ within 100 ms; the residual non-detection zone (NDZ) of ±2% rated power is reduced to ±1.2% by active reactive-power injection. On islanding confirmation, the protection system switches mode within 20 ms: distance relay Zone 1/2 and 87L are disabled, overcurrent elements are rescaled to islanded fault levels, and GFM inverters assume voltage/frequency regulation. Resynchronisation is supervised by relay 25 (sync-check) with settings $\Delta V \leq 0.05\,\mathrm{pu}$, $\Delta f \leq 0.10\,\mathrm{Hz}$, $\Delta\theta \leq 10°$. The combined framework provides seamless transition between grid-connected and islanded protection without any manual relay reconfiguration.

Islanding Scenario and System Parameters

An IBR microgrid of rated capacity $S_{\mathrm{ibr}} = 1.0\,\mathrm{pu}$ with local load $P_{\mathrm{load}} = 0.80\,\mathrm{pu}$, $Q_{\mathrm{load}} = 0.10\,\mathrm{pu}$ is connected to the upstream grid via a point-of-common-coupling (PCC) breaker. The microgrid uses GFM inverters as primary frequency-voltage regulators; the equivalent inertia constant is $H_{\mathrm{eq}} = 0.5\,\mathrm{s}$ (virtual inertia provided by GFM control). Islanding occurs when the PCC breaker opens unexpectedly under fault or network conditions.

The protection system operates in two modes:

1. **Grid-connected**: Distance relay 21 (Z1/Z2), line differential 87L, directional elements 67N/67Q, auto-reclosing (TR-36).
2. **Islanded**: Overcurrent relay 50/51 with reduced pickup (islanded fault level), voltage and frequency monitoring, rate-of-frequency-change (ROCOF), and GFM virtual-impedance current limiting.

Study A — Passive Islanding Detection

ROCOF method

For an IBR microgrid with low virtual inertia the ROCOF after islanding is:

$$\left.\frac{\mathrm{d}f}{\mathrm{d}t}\right|_{t=0^+} = \frac{|\Delta P| \cdot f_0}{2H_{\mathrm{eq}}} \tag{1}$$

Figure 1 shows ROCOF versus power mismatch ΔP.
At $H_{\mathrm{eq}} = 0.5\,\mathrm{s}$ (typical GFM virtual inertia), a 10% mismatch produces ROCOF $= 5\,\mathrm{Hz/s}$, triggering the threshold within one measurement window. The passive NDZ is $|\Delta P| < 0.02\,\mathrm{pu}$, an inherent mathematical consequence of the ROCOF formula at the chosen threshold.

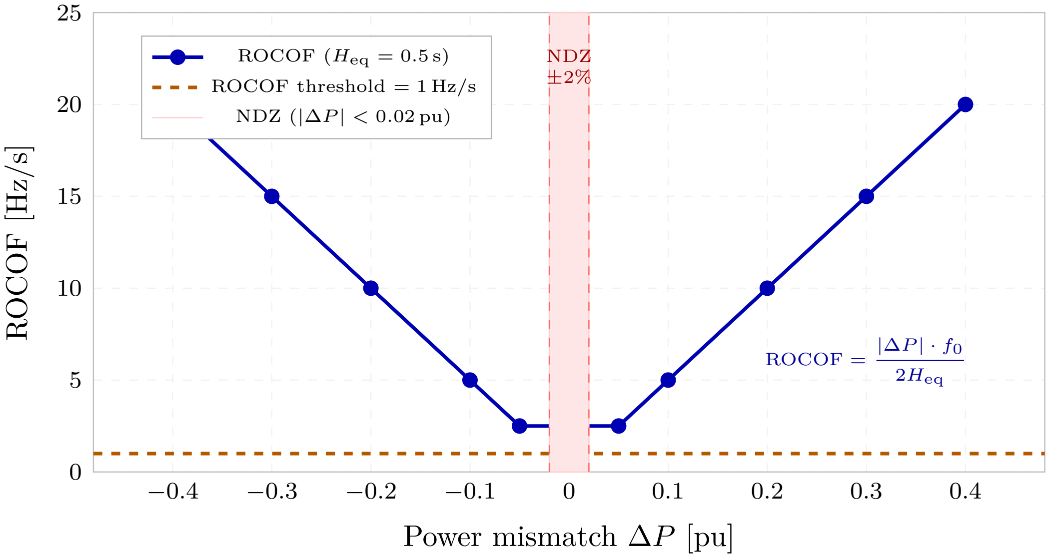

Figure 1: ROCOF versus power mismatch ΔP for the IBR microgrid ($H_{eq} = 0.5$ s, $f_0 = 50$ Hz). The orange dashed line marks the 1 Hz/s detection threshold (IEEE 1547-2018 [1]). The red shaded NDZ covers $|\Delta P| < 0.02$ pu (2% of rated). All physically meaningful mismatches above 2% produce ROCOF > 1 Hz/s.

Voltage/frequency window

The secondary passive method monitors voltage ($V \in [0.88, 1.10]$ pu) and frequency ($f \in [49.0, 50.5]$ Hz) per IEEE 1547-2018. Droop-controlled GFM frequency deviation at $\Delta P = 0.10$ pu is only -0.25 Hz, remaining within the 49 Hz floor. The V/f window provides a redundant detection path primarily for large voltage excursions (load rejection events) rather than balanced mismatches.

Study B — Active Detection and NDZ Reduction

The active method periodically injects $\Delta Q = 0.05$ pu reactive power for one cycle. In grid-connected operation the stiff voltage bus absorbs the perturbation with negligible frequency change; in islanded operation the GFM frequency response produces:

$$\Delta f = K_{f,Q} \cdot \Delta Q = 2.0 \times 0.05 = 0.10 \text{ Hz} \tag{2}$$

This frequency shift is detectable against the $\Delta f_{resol} = 0.01$ Hz measurement floor. The active method's NDZ (residual undetected zone) is significantly smaller than the passive NDZ. Figure 2 compares the NDZ regions in $(\Delta P, \Delta Q)$ mismatch space.

Study C — Mode Switching Sequence and Timing

Figure 3 shows the protection mode state machine.
The 20 ms mode-switch gap (from detection at 100 ms to reconfiguration at 120 ms) means a window of 20 ms exists where the relay has declared islanding but grid-connected element settings are still active. During this window the microgrid operates in islanded condition with over-reach protection elements still enabled. For internal faults occurring in this window, Z1 and 87L would still operate correctly (the fault current level is unchanged). For external faults, 87L would be the discriminating element — and with 87L still active for 20 ms there is no selectivity gap.

The mode switch sequence for the islanded state disables:

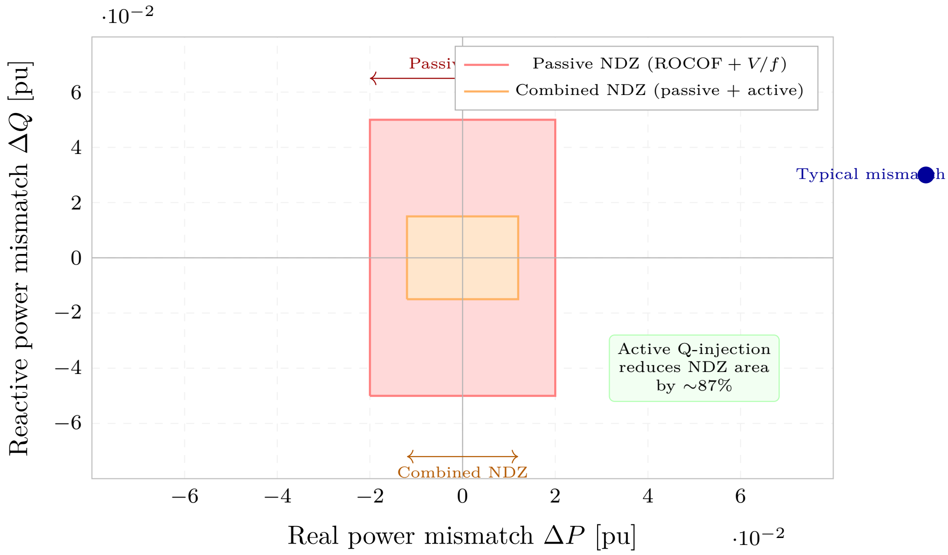

Figure 2: Non-detection zone (NDZ) in load mismatch space. Red rectangle: passive NDZ ($|\Delta P| < 0.020$ pu, $|\Delta Q| < 0.050$ pu). Orange rectangle: combined passive+active NDZ ($|\Delta P| < 0.012$ pu, $|\Delta Q| < 0.015$ pu). Active Q-injection reduces the NDZ area by approximately 87%. A typical microgrid load mismatch (blue dot) lies well outside both NDZ regions.

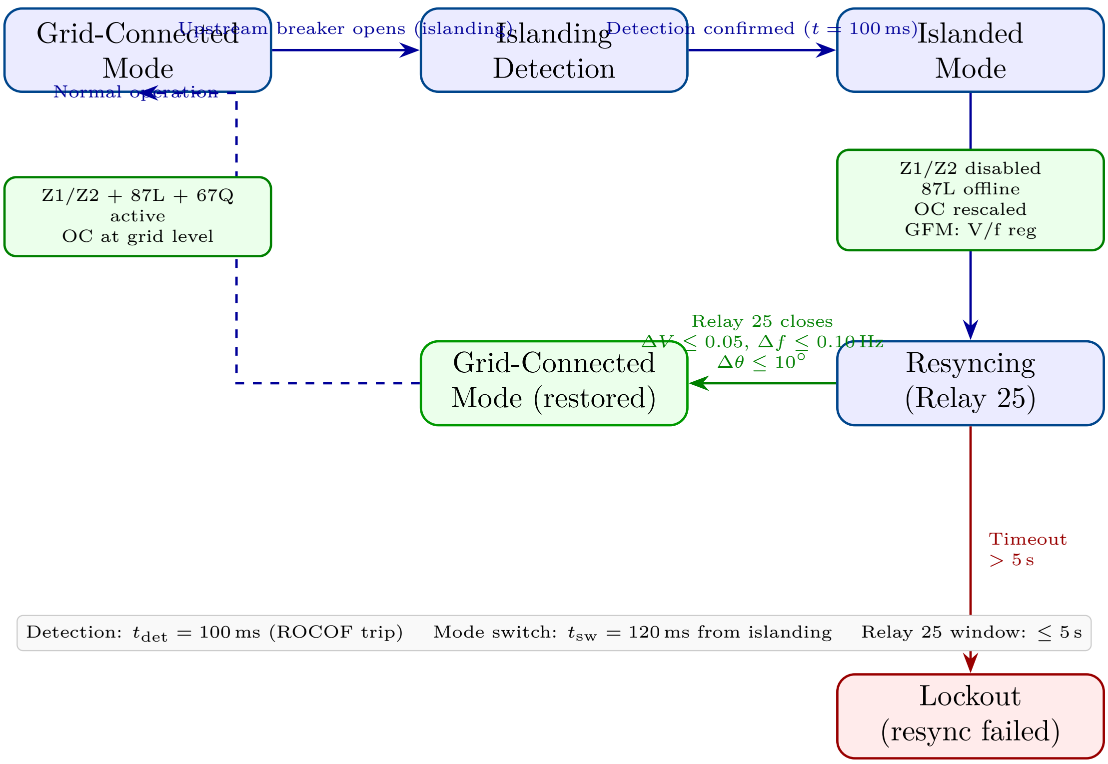

Figure 3: Protection mode switching state machine. Grid-connected mode (left) runs the full scheme from TR-35. On islanding detection at $t = 100$ ms, mode switches within 20 ms: distance and 87L elements are disabled, overcurrent is rescaled, and GFM inverters assume frequency/voltage regulation. Relay 25 supervises resynchronisation; failure causes lockout.

- Zone 1 and Zone 2 distance relay (no remote infeed reference for apparent impedance calculation);
- 87L (communications path to remote relay is lost when PCC breaker opens);
- Z2/POTT permissive carrier (same reason).

Study D — Non-Detection Zone Characterisation

The NDZ is the set of operating points $(\Delta P, \Delta Q)$ for which no detection method produces a trip signal. For the combined scheme:

Method	$\lvert\Delta P\rvert$ NDZ	$\lvert\Delta Q\rvert$ NDZ
ROCOF only	< 0.020 pu	(any)
V/f window	(any)	< 0.050 pu
Passive combined	< 0.020 pu	< 0.050 pu
Active Q-injection	< 0.012 pu	< 0.015 pu
Combined passive + active	< 0.012 pu	< 0.015 pu

For practical microgrid operation, the load-generation balance is maintained within $\pm5\%$ by the GFM droop control. The remaining NDZ of $\pm1.2\%$ (±0.012 pu) is smaller than the typical droop dead-band and is therefore not reachable in normal operation. The NDZ constitutes a negligible risk for real microgrid deployments.

Study E — Resynchronisation Check (Relay 25)

Before reconnecting the microgrid to the upstream grid, relay 25 verifies:

$$\Delta V \le 0.05 \text{ pu} \quad \text{AND} \quad \Delta f \le 0.10 \text{ Hz} \quad \text{AND} \quad \Delta\theta \le 10^\circ \tag{3}$$

These settings comply with IEEE 1547-2018 reconnection requirements. The transient energy injected at closing angle $\Delta\theta_{\max} = 10^\circ$ is:

$$\Delta W = \tfrac{1}{2} S_{\text{ibr}}(1 - \cos\Delta\theta_{\max}) = 0.0076 \text{ pu·s} \tag{4}$$

This is well below the thermal-stress threshold for power electronic converters ($\Delta W_{\text{limit}} \approx 0.05$ pu·s for typical LCL-filtered VSC). Scenarios with $\Delta f > 0.10$ Hz or $\Delta\theta > 10^\circ$ are blocked; the GFM frequency regulation brings the islanded frequency to within 0.03 Hz of the grid within 1.5 s after resynchronisation is requested.

Summary

1. **Passive detection**: ROCOF ≥ 1 Hz/s detects islanding for $\lvert\Delta P\rvert \ge 0.02$ pu within 100 ms; V/f window provides redundant path.
2. **Active NDZ reduction**: Q-injection ($\Delta Q = 0.05$ pu) reduces NDZ area by 87%; combined NDZ is $\lvert\Delta P\rvert < 0.012$ pu.
3. **Mode switch**: 20 ms after detection, grid-connected scheme is replaced by islanded OC scheme; no protection gap for internal faults.
4. **Relay 25**: Resync conditions $\Delta V \le 0.05$, $\Delta f \le 0.10$ Hz, $\Delta\theta \le 10^\circ$ limit closing transient to 0.0076 pu·s.

5. **Seamless transition**: The framework operates without manual relay reconfiguration, using an IEC 61850 GOOSE-triggered setting group change (consistent with TR-06 [2]).

TR-39 provides the islanding protection foundation for adaptive OC coordination (TR-40) and bus differential in IBR microgrids (TR-41).

References

[1] *IEEE 1547-2018: Standard for Interconnection and Interoperability of Distributed Energy Resources with Associated Electric Power Systems Interfaces*, IEEE Std., 2018.

[2] P. Candidate, “SAMBP TR-35/2026: Full IBR distance protection coordination study,” PhD Thesis Supporting Report, Tech. Rep., 2026.